

# WASP-104b

R-2499

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-104\_180324 · created 2026-08-24T23:08:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458201.8898 +/- 0.0015 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.149 +/- 0.043                 |
| Transit depth (Rp/Rs)^2                           | 2.22 +/- 1.29 [%]               |
| Orbital Inclination (inc)                         | 84.39 +/- 2.79                  |
| Airmass coefficient 1 (a1)                        | 1.11 +/- 0.51                   |
| Airmass coefficient 2 (a2)                        | -0.05 +/- 0.042                 |
| Scatter in the residuals of the lightcurve fit is | 45.34 %                         |
| Best Comparison Star                              | #3 - [302, 370]                 |
| Optimal Aperture                                  | 0.75                            |
| Optimal Annulus                                   | 3.75                            |
| Transit Duration (day)                            | 0.074 +/- 0.026                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.149 ± 0.043 vs 0.1214 (deviation 0.64σ) |
| Duration fit vs published | 0.074 d vs 0.0742 d (deviation 0.01σ)     |
| Mid-time O–C              | -3.1 min                                  |
| Depth SNR                 | 3.5                                       |
| Residual scatter          | 45.34 %                                   |

## Light curve

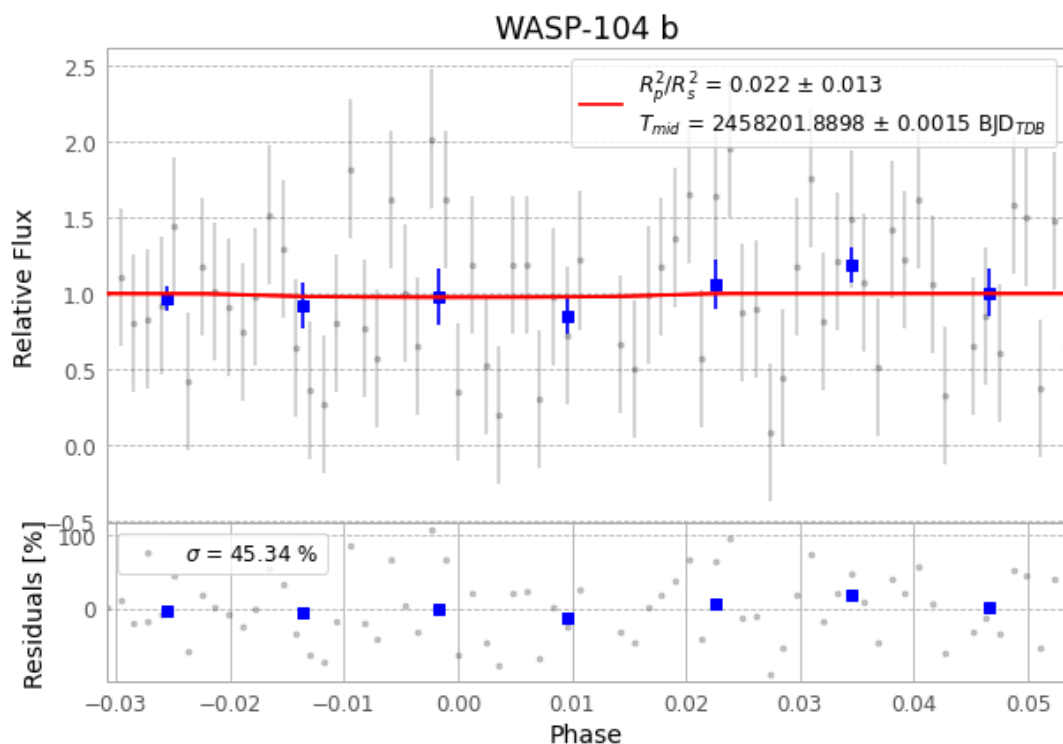


Figure 1 — FinalLightCurve\_WASP-104 b\_2018-03-24.png (SHA-256 5aab3a63a3d1df89...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 75], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0fb51d2c8eeb6693...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

72 FITS frames (47 MB), WASP-104180324075412.FITS → WASP-104180324112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180324.log (756ed99d7f0f...), FinalLightCurve\_WASP-104 b\_2018-03-24.png (5aab3a63a3d1...), FinalLightCurve\_WASP-104 b\_2018-03-24.csv (3938327454ad...)

## Compute resources

Wall time 148.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:08Z.